Experiment No. 04

Title: Exploring Software Engineering Tools JIRA, TRELLO.

Batch: C3 **Roll No.: 16014225801** **Experiment No.: 04**

Aim: To study and explore project management tools JIRA and Trello, and to understand how they support software engineering practices like task tracking, collaboration, and agile methodologies.

Resources needed:

- Internet connection
- Web browser (Chrome/Firefox)
- JIRA software (Cloud/Server edition)
- Trello (Cloud version)

Theory:

1. **JIRA** - **(**Developed by Atlassian)
   - JIRA is primarily used for **issue tracking, bug tracking, and agile project management**.
   - Features:
     - Scrum and Kanban boards
     - Sprint planning and tracking
     - Advanced reporting (burndown charts, velocity charts)
     - Integration with CI/CD tools (Bitbucket, GitHub, Jenkins)

2. **Trello**
   - Visual project management tool using **Kanban boards**.
   - Features:
     - Cards and lists for task organization
     - Drag-and-drop workflow
     - Power-ups (integrations with Slack, GitHub, Calendar)
     - Suitable for small to medium-sized projects and team collaboration

Comparison

- JIRA is more suitable for **complex enterprise projects** with multiple teams.
- Trello is simple, intuitive, and suited for **lightweight project/task management**.

Activities:

1. Create an account in **JIRA** and **Trello**.

2. In JIRA:
   - Create a project.
   - Configure Scrum/Kanban board.
   - Add issues (stories, tasks, bugs).
   - Assign to team members and track progress.

3. In Trello:
   - Create a new board.

- Add lists (To Do, In Progress, Done).
- Add cards for tasks, assign members, set deadlines.
- Enable calendar power-up to visualize timelines.

4. Compare features of JIRA and Trello in terms of **usability, scalability, and reporting**.

Step-by-Step Guide: JIRA

1. Sign Up

- Go to https://www.atlassian.com/software/jira/free
- Click **Get it free** → Sign up using Google/Microsoft/email.
- Confirm email → You'll be redirected to **JIRA dashboard**.

2. Create a Project

- On the left menu, click **Projects → Create Project**.
- Choose template:
  - **Scrum** → Sprint-based planning.
  - **Kanban** → Continuous workflow.
- Enter project name (e.g., *SE-Tools-Experiment*).
- JIRA automatically creates a **board**.

3. Add Issues (Tasks)

- On your project board, click **Create Issue**.
- Fill fields:
  - *Type*: Story, Task, Bug.
  - *Summary*: Short title (e.g., "Implement login feature").
  - *Assignee*: Select team member.
- Issues will appear as **cards** on board.

4. Use the Board

- Board columns = workflow stages (**To Do → In Progress → Done**).
- Drag and drop issues across columns as they progress.
- Scrum projects also include a **Backlog** for sprint planning.

5. Reports

- From the side menu, open **Reports**.
- View **Burndown Chart**, **Velocity Report**, and **Cumulative Flow Diagram**.
- These help analyze productivity and sprint performance.

JIRA Lab Task:

Create a Scrum project → Add 3 tasks → Assign to team members → Move across workflow → View Burndown Chart.

Step-by-Step Guide: Trello

1. Sign Up

- Go to https://trello.com/signup
- Sign up with Google/email.
- You'll land on **Trello Workspace**.

2. Create a Board

- Click **Create new board**.
- Give it a name (*SE-Tools-Trello-Board*).
- Choose background (optional).
- Board opens with empty canvas.

3. Add Lists

- Create 3 lists (columns):
  - **To Do**
  - **In Progress**
  - **Done**

4. Add Cards (Tasks)

- Under **To Do**, click **+ Add a card**.
- Enter task name (e.g., "Write SRS Document").
- Click card → Add details: description, checklist, labels, due date, assignees.

5. Manage Workflow

- Drag cards between lists (To Do → In Progress → Done).
- Cards update visually so the whole team can track progress.

6. Use Power-Ups

- Click **Power-Ups** on board → Add **Calendar**.
- Cards with due dates now appear in calendar view.
- Enhances project planning.

Trello Lab Task:

Create a Kanban board → Add 5 tasks as cards → Assign deadlines → Enable Calendar power-up → Track workflow visually.

Questions:

1. What advantages do agile tools like JIRA and Trello provide compared to traditional project management methods?
2. How does JIRA's sprint tracking differ from Trello's Kanban-style task management?
3. Which scenarios are more suitable for Trello than JIRA, and why?

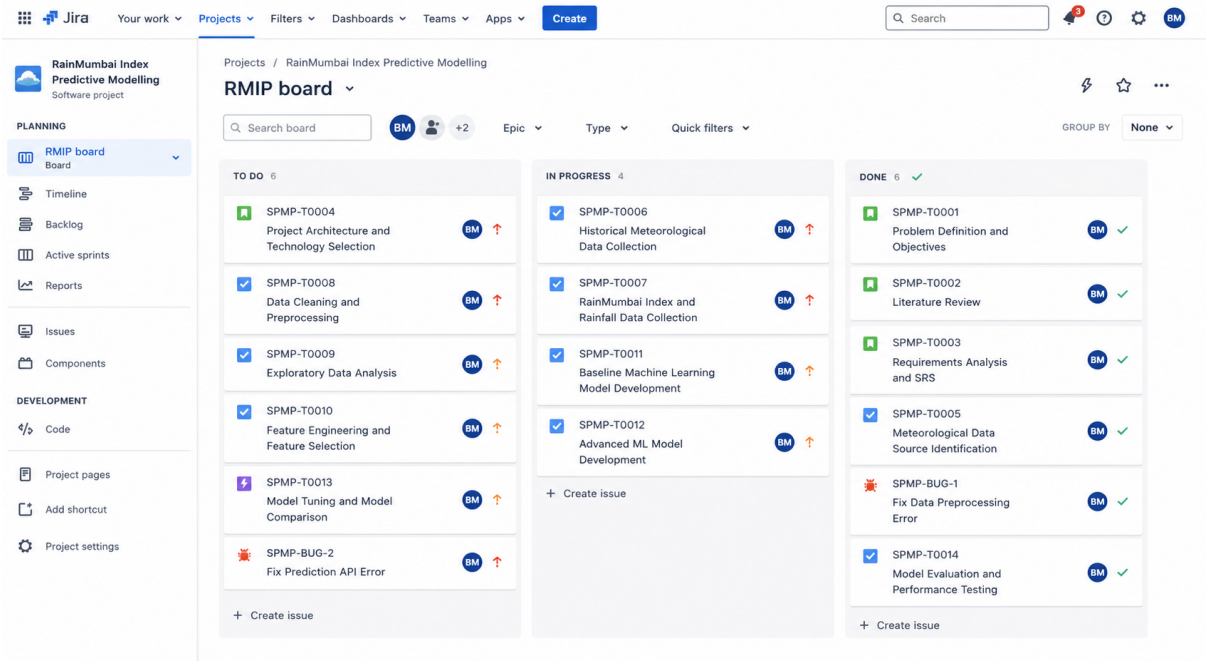
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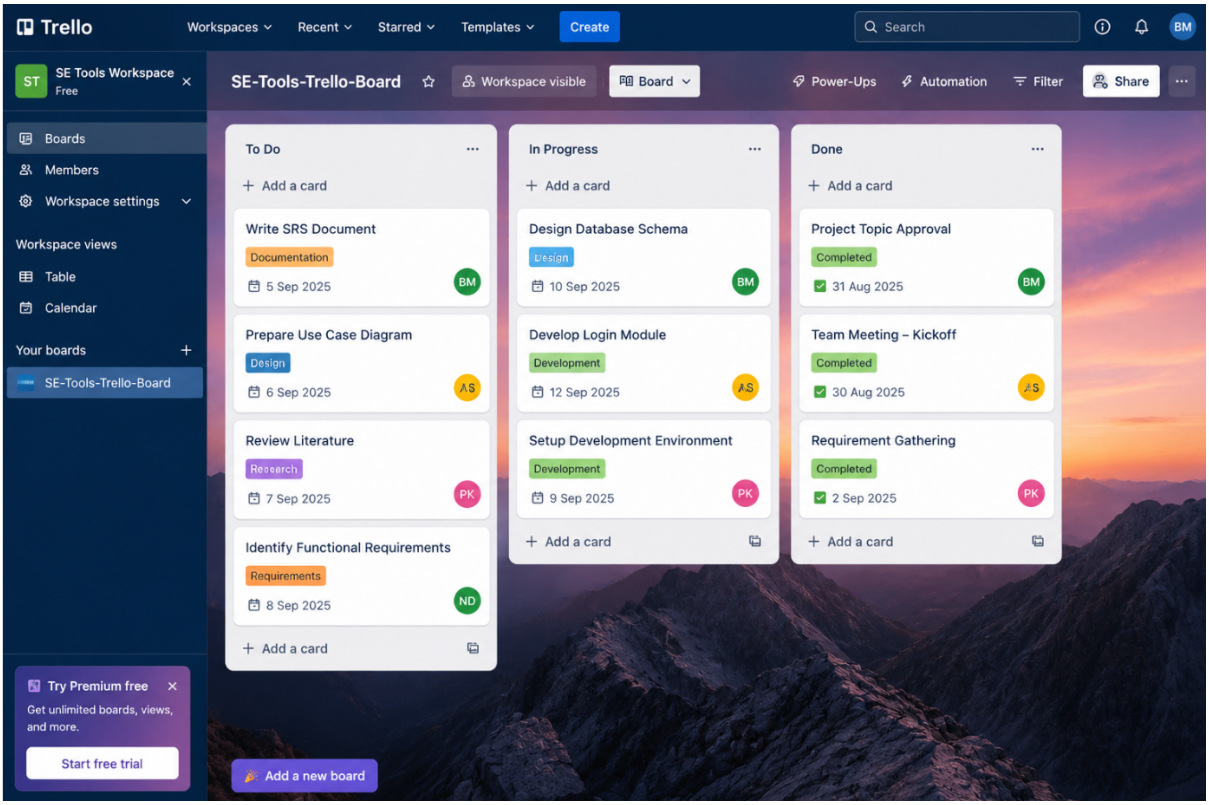
1. Agile tools like JIRA and Trello provide better flexibility, collaboration, and transparency than traditional methods. They allow teams to track tasks, update progress, and manage changes in real time. They also improve communication and make project progress easier to monitor.

2. JIRA's sprint tracking focuses on time-bound iterations with sprint goals, backlogs, and progress tracking. Trello uses a Kanban-style board where tasks move through stages such as To Do, Doing, and Done. JIRA is better for structured Agile development, while Trello is simpler and more visual.

3. Trello is more suitable for small projects, personal tasks, event planning, and simple team workflows. It provides an easy-to-use visual board without the complexity of sprints and detailed issue tracking. Therefore, Trello is preferred when simplicity, quick setup, and flexible task management are important.

Results:





Conclusion:

Hence we Performed the experiment successfully.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

Books/ Journals/ Websites:

1. Roger S. Pressman, Software Engineering: A practitioners Approach, 7th Edition, McGraw Hill, 2010
2. https://www.sharelatex.com/
3. Atlassian Documentation – JIRA Software Guide
4. Trello Help Center – trello.com/guide